

Jisu Shin

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RESEARCH INTERESTS

✓ **3D Vision and Digital Twin Systems** – Human & Scene Reconstruction and Interaction, 3D Gaussian Splatting, 3D Object Pose Tracking

✓ **Embodied Perception and Physical AI** – Physically Interactive Avatars, Vision-based Perception for Robotics

✓ **Computational Photography** – Camera ISP Modeling, Image Restoration, Image Harmonization

EDUCATION

Mar. 2021 ~ **Gwangju Institute of Science and Technology** Gwangju, Korea
Artificial Intelligence Graduate School
M.S./Ph.D. Integrated Program(Ph.D.)
Expected Graduation: **February 2027**

Current research location since Sep. 2025~: **Yonsei University (Advisor Relocation)**

Jun. 2018 ~ **Exchange Program on UC Berkeley** University of California, Berkeley
Aug. 2018 Discrete Mathematics, Organic Chemistry Course

Mar. 2017 ~ **Gwangju Institute of Science and Technology** Gwangju, Korea
Feb. 2021 Department of Electrical Engineering and Computer Science
B.S. in Electronic Engineering and Computer Science

RESEARCH EXPERIENCE

- **Research Intern** at Huawei Noah's Ark Lab 3D Vision Team, United Kingdom, Dec. 2024 ~ Sep. 2025
✓ 3D scene reconstruction using Gaussian Splatting under varying illumination and Camera ISP conditions
- **Research Intern** at *Korea Electronics Technology Institute (KETI)*, South Korea Jul. 2023 ~ Feb. 2024
✓ Reconstructing a realistic drivable 3D human model from sequential images
- **Research Intern** at *Electronics and Telecommunications Research Institute (ETRI)*, South Korea
Jul. 2019 ~ Aug. 2019
✓ Research on Wireless Communication using Raspberry Pi

PUBLICATIONS

- **J.S. Shin**, J.O. Lee, J.G. Lee, I.H. Bae, D.H. Lee, H.K. Im, Y.W. Lee, H.G. Jeon, “ComPose: When to Trust Hands for Object Pose Tracking” *Neural Information Processing Systems (NeurIPS)*, Dec 2026
- **J.S. Shin**, S.H. Shin, H.G. Jeon, “Scene Structure Guidance Network: Unfolding Graph Partitioning into Spatial Feature Learning” Under Review
- S.H. Han, M.G. Park, **J.S. Shin**, S.H. Shin, J.H. Park, H.G. Jeon, “PIAvatar: Physically Interactive Avatars via Deformation Gradient Decoupling” *European Conference on Computer Vision (ECCV)*, Sep 2026
- J.G. Cho, Y.J. Park, S.J. Ha, S.Y. Kim, S.H. Park, **J.S. Shin**, J.M. Lee, C.M. Hwang, H.J. Jung, S.E. Lee, H.G. Jeon, “From Script to Shot: A Benchmark for Grounding Screenplays in Movies” *European Conference on Computer Vision (ECCV)*, Sep 2026
- J.O. Lee, J.M. Lee, Y.J. Song, I.H. Bae, **J.S. Shin**, H.G. Jeon, J.H. Kim, “Relaxed Rigidity with Ray-based Grouping for Dynamic Gaussian Splatting” *European Conference on Computer Vision (ECCV)*, Sep 2026
- **J.S. Shin**, Richard Shaw, S.H. Shin, Zhensong Zhang, H.G. Jeon, Eduardo Pérez-Pellitero, “CHROMA: Consistent Harmonization of Multi-View Appearance via Bilateral Grid Prediction” *International Conference on Learning Representations (ICLR)*, Apr 2026
- Arthur Moreau, Richard Shaw, Michal Nazarczuk, **J.S. Shin**, Thomas Tanay, Zhensong Zhang, Songcen Xu, Eduardo Pérez-Pellitero, “Off the Grid: Detection of Primitives for Feed-Forward 3D Gaussian Splatting” *In Proc. of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, Jun 2026
- S.H. Shin, D.M. Shin, **J.S. Shin**, H.G. Jeon, J.Y. Lee, “Video Color Grading via Look-Up Table Generation” *International Conference on Computer Vision (ICCV)*, Oct 2025
- **J.S. Shin**, J.M. Lee, S.M. Lee, M.G. Park, J.M. Kang, J.H. Yoon, H.G. Jeon, “CanonicalFusion: Generating Drivable 3D Human Avatars from Multiple Images” *European Conference on Computer Vision (ECCV)*, Sep 2024
- M.B. Kim, M.S. Jeong, M.Y. Lee, J.H. Kim, Y.J. Choi, S.S. Kim, H.G. Jeon, **J.S. Shin**, “The path-tracing simulation of light-field camera system: SurfCam/GrainCams for lunar surface exploration” *Advances in Space Research*, 2024
- S.H. Shin, **J.S. Shin**, J.H. Bae, I.W. Shim, H.G. Jeon, “Close Imitation of Expert Retouching for Black-and-White Photography” *In Proc. of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, Jun 2024
- **J.S. Shin**, S.H. Shin, H.G. Jeon, “Task-specific Scene Structure Representations.” *In Proc. of Association for the Advancement of Artificial Intelligence (AAAI)*, Feb 2023, [Oral Presentation]

HONORS & AWARDS

- Selected for the Doctoral Colloquium and received the Best Presentation Award, Korean Conference on Computer Vision (KCCV), Aug 2026
- Selected as Outstanding Reviewer, European Conference on Computer Vision (ECCV), Jul 2026
- Travel Scholarship Funding - ICLR-26 Student Scholarship, Rio de Janeiro, Brazil, Apr 2026
- Scholarship Funding - RA Student Research Achievement Scholarship, AI Graduate School, GIST, Dec 2025
- Selected as the Participants for International Computer Vision Summer School (ICVSS), July 2025
- Selected as Internship Program on 3D Vision Team at Huawei Noah's Ark Lab, United Kingdom, Dec 2024 ~ Sep 2025
- Finalist, 2024 Qualcomm Innovation Fellowship Korea, Qualcomm Technologies, Inc., Dec 2024
- 7th Hong Bok-Soon Scholarship, Development Foundation, GIST, Nov 2024.

- Selected as Internship Program on Korea Electronics Technology Institute (KETI), Jun 2023 ~ Mar 2024
- Travel Scholarship Funding - AAAI-23 Student Scholarship, Washington, DC, USA, Feb 2023
- Excellent Paper Award, The 29th International Workshop on Frontiers of Computer Vision (IW-FCV 2023), Feb 2023
- Selected as Internship Program on Electronics and Telecommunications Research Institute (ETRI), Jul 2019 ~ Aug 2019
- Selected as Summer School Exchange Program Scholarship on UC Berkeley, Jun 2018 ~ Aug 2018

PATENTS

- H.G. JEON, **J.S. SHIN**, S.H. SHIN, “IMAGE HARMONIZATION METHOD THROUGH INPAINTING BOUNDARIES OF FOREGROUND IMAGE”
Korea – Patent No.10-2562677
- H.G. JEON, **J.S. SHIN**, S.H. SHIN, “METHOD FOR GENERATING TASK-SPECIFIC SCENE STRUCTURE” U.S. – Application No.18/410,525
- H.G. JEON, S.H. SHIN, **J.S. SHIN**, J.H. BAE, I.W. SHIM, “METHOD FOR CREATING BLACK-AND-WHITE PHOTO THAT IMITATES EXPERT RETOUCHING STYLES” U.S. – Application No.18/508,456
- H.G. JEON, **S.H. SHIN**, J.S. SHIN, “METHOD FOR GENERATING TASK-SPECIFIC SCENE STRUCTURE” Korea – Application No.10-2023-0008665
- H.G. JEON, **S.H. SHIN**, J.S. SHIN, J.H. BAE, I.W. SHIM, “METHOD FOR CREATING BLACK-AND-WHITE PHOTO THAT IMITATES EXPERT RETOUCHING STYLES” Korea – Application No.10-2023-0030667